RUSHIL BARVE

Computer Science & Computational Mathematics · University of Waterloo



About Me

First-year Waterloo student who builds practical, privacy-minded software, competes in hackathons and olympiads, and leads teams. Looking for Winter/Spring co-op.

Contact

EMAIL
s2rkv0110@gmail.com
PORTFOLIO
rsk0110.github.io/s2rkportfolio
PHONE
+1 548 384 6520
LINKEDIN
linkedin.com/in/rushil0110
GITHUB
github.com/Rsk0110
DEVPOST
devpost.com/Rsk0110
LOCATION
Waterloo, Ontario

Skills

LANGUAGES
Python, JavaScript, TypeScript, GDScript, C, SQL, HTML, CSS
FRAMEWORKS
React, Vite, Tailwind CSS, FastAPI, Godot 4, Zustand
TOOLS
Git, GitHub Actions, SQLite, IndexedDB, Render

Achievements

- Gold Medal of Distinction, International Informatics Olympiad (2023-24); Bronze (2024-25)
- Gold Medal of Distinction, National Science Olympiad
- Six Best Delegate awards across 8+ national and international MUN conferences
- Basketball, CBSE Qatar Clusters: Gold (2025-26), Bronze (2023-24, 2024-25)

Waterloo Teams

- WARG (Waterloo Aerial Robotics): Autonomy and Embedded Flight Software teams
- UW Orbital: student satellite design team
- UW Game Dev · Computer Science Club · UW Data Science Club

EDUCATION

University of Waterloo — Expected 2031
Computer Science (Co-op) · Computational Mathematics

Delhi Public School – Modern Indian School, Doha — Graduated 2026
CBSE Senior Secondary, Computer Science

PROJECTS

CircuitGuard — Python · FastAPI · React · Three.js
Winner, Ampere – AI for Motor Safety · FormulaTech Hacks 2026

- Real-time Formula 1 safety system: analyses live driver telemetry to detect crashes and stopped cars on the racing line and recommends the FIA response (safety car, VSC, red flag) under the 2026 sporting regulations.
- Models per-sector spectator risk with a Bayesian Monte Carlo insurance-pricing model, so premiums can reflect each track section instead of a blanket rate.
- Python/FastAPI backend on the OpenF1 API, FAISS semantic search over FIA regulations, faster-whisper radio transcription, Next.js/React 3D track view.

Whisper Chain — Godot 4 · GDScript
Puzzle game built on function composition

- Puzzle game where each villager garbles a message (reverse, swap, double, drop a word); players route the message so the distortions cancel, discovering inverses and non-commutativity.
- Graph solver (DFS over simple paths) finds every correct route, sets par and powers live hints; used to design and validate all 10 levels.
- ~3,000 lines of GDScript with UI-free game logic, ~1,500 test assertions including randomized property tests, and GitHub Actions CI building web and Windows exports.

MUN Chair Toolkit — TypeScript · React · Vite · Tailwind · Zustand · Dexie
Local-first web app for Model UN chairs · live on GitHub Pages

- Committee-management app for Model UN chairs: committee and agenda setup, bulk delegation entry, and roll call with Present, Present and Voting, and Absent states.
- Automated quorum, simple-majority and two-thirds calculations, so chairs no longer count votes by hand.
- Kept rules in framework-free functions (src/lib/rules.ts) with automated tests; deployed to GitHub Pages.

Door Occupancy Counter (in progress) — Python · SQLite · JavaScript
Prototype: live building occupancy from door sensors

- Counts people entering and leaving each door from two-beam sensor events (direction from beam order), keeping a live headcount with no manual logging.
- Live web dashboard, SQLite event log and automatic daily Excel reports (summary, hourly and event sheets) written with no external libraries.

Source code available **on request** · more projects at rsk0110.github.io/s2rkportfolio

LEADERSHIP & EXPERIENCE

Director of MUN & Electoral and Democratic Club — May 2025 – Jun 2026
DPS-Modern Indian School, Doha

- Led and managed the school's MUN program: organized conferences, trained delegates, promoted diplomacy and public speaking.

Head of Marketing, TEDxYouth@DPSMIS — Sep – Dec 2025

- Led branding, social media and promotion for a TEDx-licensed youth event, working across curation, design, operations and technical teams.

Head of Operations, DPSDQ MUN — Jun – Oct 2025

- Ran logistics and delegate services for the 25th-anniversary edition, coordinating with the Secretariat.

Team Leader, Qatar Children's Science Congress — Dec 2024 – Mar 2025

- Led an original student research project from experimental design to data analysis and expert-panel presentation.

HACKATHONS

FormulaTech Hacks — Winner (Ampere)	2026	RevenueCat Shipaton	2026
OpenCV AI Competition (AWS)	2026	Nebius x NVIDIA Global AI Hackathon	2026
IBM BOB 2.0 Hackathon	2026	DPS MIS Doha Hackathon	2025
Birla School Hackathon — Qatar	2024	CMU Qatar Hackathon	2023

8 hackathons · 5 won · 1 organised